Introduction

Along these tutorials, we will see how **SCOPE** interacts with the different parts of the code to handle the execution of computational workflows.

These are the topics covered in each tutorial:

1. The **System** class and its sources: the **Specie**, **Cell** and **Atom** classes
2. The Computational workflow: **Branch**, **Workflow**, **Job**, and **Computation** classes
3. The **State** class
4. The **Data**, **Collection** and **VNM** classes
5. The **Input_data** class, and **scope input files**
6. Running **SCOPE** - Part 1: File Structure
7. Running **SCOPE** - Part 2: Execution
8. Running **SCOPE** - Part 3: Detailed Actions
9. The Azo add-on: Subclasses, Analysis and Creation of Azo Systems

Tutorial 2: The Computational Workflow

SCOPE aims at automatizing computational workflows, or tasks. To encode the potential complexity of such workflows, it needs a flexible hierarchy of classes, whose purpose might not be evident to a new user.

As currently implemented, the hierarchy of classes associated with a computational workflow goes like this: **Branch** -> **Workflow** -> **Job** -> **Computation**.

From Bottom to Top:

- **COMPUTATION**: Starting from the bottom, a **Computation** is associated with an input, an output and a submission file, and contains the information of this single **Computation** that is **submitted** to a computational cluster. Whether it worked or not, SCOPE takes the results into account to decide the course of action.
- **JOB**: A **Job** is a single, or a collection of **Computations** with a specific goal. For instance, a **Job** could be the Single Point Energy evaluation of a given structure. It might work with only one computation, or it might need more than one, for instance if convergence issues arise.

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- **WORKFLOW**: A collection of **Jobs** for a given molecule or unit cell in a system (i.e a source) form a **Workflow**. For instance, a **Workflow** could be: (1) Optimization with Cheap DFT Functional, (2) Optimization with Expensive DFT Functional, (3) Hessian evaluation (frequencies). Workflows are connected to the **Source** of a **System**
- **BRANCH**: A **Branch** is a collection of related **Workflows**. For instance, the Workflows for the high-spin and low-spin versions of the same molecule can be put together under the same **Branch**.

In this tutorial, we will open an existing **SYSTEM**, stored in a binary file, and navigate the computational workflow inside. Also, we will see how to create a computational workflow manually

In [1]:
```
import os
import scope
```

In [2]:
```
## Path of the data folder. It should be "os.path.abspath('.')+'/Data"
data_folder = os.path.abspath('../')+'/Data/1-Tutorial_1/'

## Loads the System object from a binary file, provided in the tutorial folc
sys = scope.load_binary(f"{data_folder}/Systems/ABITEM/ABITEM.npy")

## This will set the main path of the system to the data folder, so that any
sys.set_main_path(data_folder)
```

Out[2]:
```
True
```

In [3]:
```
## All objects in SCOPE have a __repr__ method, so printing shows a summary
print(sys)
```

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```
-------------------------------------
-- >>> SCOPE System_sco Object >>> --
-------------------------------------
 Version                = 1.0
 Type                   = system
 Subtype                = sco_system
 Source Path            = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources            = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches           = 1
     0: isolated, Num Workflows: 2
```

In [4]:
```python
## Shows the branches of a system
print(sys.branches)
```

```
[-----------------------------------------------------
   >>> BRANCH
-----------------------------------------------------
 System                 = ABITEM
-----------------------------------------------------
 self.status            = active
 self.creation_time     = 13/03/2026 16:50:06
 self.creation_user     = svela
 self.path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
 self.name              = isolated
 Num Workflows          = 2

]
```

Branch

In [5]:
```python
## Select a given branch either by selecting it from the list, or searching
found, this_branch = sys.find_branch("Isolated")
print(found)
print(this_branch)
```

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```
## You'll likely see a WARNING. This is because the branch was created in my
## You can safely ignore this WARNING for the moment.
```

```
True
----------------------------------------------------
   >>> BRANCH
----------------------------------------------------
 System                = ABITEM
----------------------------------------------------
 self.status           = active
 self.creation_time    = 13/03/2026 16:50:06
 self.creation_user    = svela
 self.path             = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
 self.name             = isolated
 Num Workflows         = 2
```

In [6]:
```
## Workflows are easily accessible. Here there are two workflows, one for th
this_branch.workflows
```

Out[6]:
```
[----------------------------------------------------
    >>> >>> WORKFLOW
  ----------------------------------------------------
   Source Name                 = ref_hs_mol
   Source Type                 = specie
   Source sub-Type             = molecule
   Source Charge               = 0
   Source Spin                 = 4
  ----------------------------------------------------
   Workflow Name               = ref_hs_mol
   Num Jobs                    = 3
         Last Job Name        = freq
         Last Job Hierarchy   = 3
  ,
  ----------------------------------------------------
    >>> >>> WORKFLOW
  ----------------------------------------------------
   Source Name                 = ref_ls_mol
   Source Type                 = specie
   Source sub-Type             = molecule
   Source Charge               = 0
   Source Spin                 = 0
  ----------------------------------------------------
   Workflow Name               = ref_ls_mol
   Num Jobs                    = 3
         Last Job Name        = freq
         Last Job Hierarchy   = 3
  ]
```

Workflow

In [7]:
```
## Again, you can select a given workflow either by selecting it from the li
this_workflow = this_branch.find_workflow("ref_hs_mol")
```

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```
print(found)
print(this_workflow)
```

```
True
----------------------------------------------------
   >>> >>> WORKFLOW
----------------------------------------------------
 Source Name                 = ref_hs_mol
 Source Type                 = specie
 Source sub-Type             = molecule
 Source Charge               = 0
 Source Spin                 = 4
----------------------------------------------------
 Workflow Name               = ref_hs_mol
 Num Jobs                    = 3
        Last Job Name        = freq
        Last Job Hierarchy   = 3
```

In [8]:
```
## This workflow has 3 Jobs, the latest being a b3lyp frequency computation
this_workflow.jobs
```

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Out[8]:
```
[-----------------------------------------------------
   >>> >>> >>> JOB
 -----------------------------------------------------
  Source Type           = specie
  Source Name           = ref_hs_mol
  Branch Name           = isolated
  Workflow Name         = ref_hs_mol
 -----------------------------------------------------
  Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tu
torials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
  Job name              = opt
  Job type              = opt
  Job hierarchy         = 1
  Job requisites        = []
  Job constrains        = ['self']
  Job setup             = regular
  Num Computations      = 1
 -------------------------------------------------------
  self.isregistered (Temp) = True
  self.isgood       (Temp) = True
  self.isfinished   (Temp) = True
 ,
 -----------------------------------------------------
   >>> >>> >>> JOB
 -----------------------------------------------------
  Source Type           = specie
  Source Name           = ref_hs_mol
  Branch Name           = isolated
  Workflow Name         = ref_hs_mol
 -----------------------------------------------------
  Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tu
torials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
  Job name              = opt2
  Job type              = opt
  Job hierarchy         = 2
  Job requisites        = ['opt']
  Job constrains        = ['self']
  Job setup             = regular
  Num Computations      = 1
 -------------------------------------------------------
  self.isregistered (Temp) = True
  self.isgood       (Temp) = True
  self.isfinished   (Temp) = True
 ,
 -----------------------------------------------------
   >>> >>> >>> JOB
 -----------------------------------------------------
  Source Type           = specie
  Source Name           = ref_hs_mol
  Branch Name           = isolated
  Workflow Name         = ref_hs_mol
 -----------------------------------------------------
  Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tu
torials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
                        = freq
  Job type              = freq
```

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```
 Job hierarchy         = 3
 Job requisites        = ['opt2']
 Job constrains        = ['self']
 Job setup             = regular
 Num Computations      = 1
-----------------------------------------------------
 self.isregistered (Temp) = True
 self.isgood       (Temp) = True
 self.isfinished   (Temp) = True
]
```

Job

In [9]:
```
## Again, you can select a given JOB either by selecting it from the list, c
found, this_job = this_workflow.find_job('opt2') ## opt2 is the name given t
print(found)
print(this_job)

## Alternatively, you can also search by hierarchy number:
found, that_job = this_workflow.find_job(hierarchy=2) ## hierarchy goes from
print(found)
print(that_job)
```

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```
True
-----------------------------------------------------
   >>> >>> >>> JOB
-----------------------------------------------------
 Source Type           = specie
 Source Name           = ref_hs_mol
 Branch Name           = isolated
 Workflow Name         = ref_hs_mol
-----------------------------------------------------
 Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
 Job name              = opt2
 Job type              = opt
 Job hierarchy         = 2
 Job requisites        = ['opt']
 Job constrains        = ['self']
 Job setup             = regular
 Num Computations      = 1
-------------------------------------------------------
 self.isregistered (Temp) = True
 self.isgood       (Temp) = True
 self.isfinished   (Temp) = True


True
-----------------------------------------------------
   >>> >>> >>> JOB
-----------------------------------------------------
 Source Type           = specie
 Source Name           = ref_hs_mol
 Branch Name           = isolated
 Workflow Name         = ref_hs_mol
-----------------------------------------------------
 Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/
 Job name              = opt2
 Job type              = opt
 Job hierarchy         = 2
 Job requisites        = ['opt']
 Job constrains        = ['self']
 Job setup             = regular
 Num Computations      = 1
-------------------------------------------------------
 self.isregistered (Temp) = True
 self.isgood       (Temp) = True
 self.isfinished   (Temp) = True
```

In [10]:
```
## Both ways you retrieve the same job.
this_job == that_job
```

Out[10]: True

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```
that jobs have requisites and constrains. This means that, when su
print(this_job.requisites)
```

```
print(this_job.constrains)
```

```
['opt']
['self']
```

In [12]:
```
## Jobs store the information provided by the user here:
print(this_job.job_data)
## Some entries inside job_data create an attribute in the object, so you ca
print(this_job.istate, '-->', this_job.job_data.istate)
```

```
Formatted input interpretation: ( self -> Instance of class Input() )
-----------------------------------------------------
self.Key                  | Data Type           | Value
-----------------------------------------------------
self.branch               | <class 'str'>       | isolated
self.workflow             | <class 'list'>      | ['ref_hs_mol', 'ref_ls_mo
l']
self.job                  | <class 'str'>       | opt2
self.istate               | <class 'str'>       | pbe_opt
self.fstate               | <class 'str'>       | b3lyp_opt
self.hierarchy            | <class 'int'>       | 2
self.requisites           | <class 'list'>      | ['opt']
self.constrains           | <class 'list'>      | ['self']
self.must_be_good         | <class 'bool'>      | True
self.job_setup            | <class 'str'>       | regular
self.job_type             | <class 'str'>       | opt

pbe_opt --> pbe_opt
```

In [13]:
```
## Some of this information will be discussed below when talking about the c
print(this_job.istate)
print(this_job.fstate)
## States are a way to manage the results of computations. They are discusse
## For the moment, suffices to say that computations take the information fr

## In this case, the job reads the data from the PBE_OPT state (e.g. initial
## Because B3LYP_OPT state does not exist when submitting the computation, t
```

```
pbe_opt
b3lyp_opt
```

Computation

In [14]:
```
## You can access the computations by selecting it from the list, or searchi
found, this_comp = this_job.find_computation(step=1, run_number=1) ## both s
print(found)
print(this_comp)
```

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```
True
--------------------------------------------------
   >>> >>> >>> >>> COMPUTATION
--------------------------------------------------
 Source Type           = specie
 Source sub-Type       = molecule
 Branch Name           = isolated
 Workflow Name         = ref_hs_mol
 Job Name              = opt2
 Job Type              = opt
--------------------------------------------------
 Initial State         = pbe_opt
 Final State           = b3lyp_opt
 Comp software         = g16
 Comp index            = 1
 Comp step             = 1
 Comp run_number       = 1
 Comp inp_path         = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/ABITEM_ref_hs_mol_opt2_
r1.com
 Comp out_path         = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/ABITEM_ref_hs_mol_opt2_
r1.log
 Comp isregistered     = True
 Comp isgood           = True
 Comp isfinished       = True
 Comp elapsed_time     = 44218.6 seconds
 Has OUTPUT            = NO
```

In [15]:
```
## If everything goes well, most jobs will be completed after a single compu
print(this_comp.step, this_comp.run_number, this_comp.isgood)
## Larger values for "run_number" are typical if a computation fails to comp
## In such cases multiple computations with increasing run_number values wil
```

```
1 1 True
```

In [16]:
```
## There is only one type of job setup for which "step" can be different tha
print(this_job.job_setup)

## When .setup is "rep_opt", it will run several 'external' steps of geometr
## With Quantum Espresso, variable-cell geo-opts computations can finish suc
## However, if you re-submit a new computation to continue the previous one,
```

```
regular
```

In [17]:
```
## The paths of the input and output files are stored. The name of these fil
this_comp.filename
```

Out[17]:
```
sys_name: ABITEM, Format: ABITEM
sou_name: ref_hs_mol, Format: ref_hs_mol
job_name: opt2, Format: opt2
run_number: 1, Format: r1
```

In [18]:
```
## The Quantum Chemistry parameters of this computation are stored in .qc_da
this_comp.qc_data
```

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Out[18]:
```
Formatted input interpretation: ( self -> Instance of class Input() )
-----------------------------------------------------
self.Key                  | Data Type           | Value
-----------------------------------------------------
self.software             | <class 'str'>       | g16
self.comp_type            | <class 'str'>       | opt
self.functional           | <class 'str'>       | b3lyp**
self.basis                | <class 'str'>       | def2svp
self.is_grimme            | <class 'bool'>      | True
self.loose_opt            | <class 'bool'>      | False
self.tight_opt            | <class 'bool'>      | False
self.grimme_type          | <class 'str'>       | d2
self.ultrafine_grid       | <class 'bool'>      | False
self.fctype               | <class 'str'>       | recalcfc
self.recalc_steps         | <class 'int'>       | 30
self.fstate               | <class 'str'>       | b3lyp_opt
self.istate               | <class 'str'>       | pbe_opt
```

In [19]:
```
## Some of this parameters are defined by the used in the job_file that spec
## Basically, it adds some defaults.
```

In [20]:
```
## Once finished, the computation is registered. This means that the output
## This is done through the "OUTPUT" class. There is one output class for ea
## When parsing outputs, an output class will read the lines of the output f
## Once finished, the lines of the output file are removed from memory. So w

## Now we do it by just creating the OUTPUT object:
output = this_comp.create_output()
print(output)
```

```
-----------------------------------------------------
          SCOPE Gaussian16 OUTPUT CLASS
-----------------------------------------------------
 Num of Lines     = 67424
 Computation Type = opt
 Requisites       = ['scf', 'opt']
-----------------------------------------------------
```

In [21]:
```
import matplotlib.pyplot as plt

## And now we can parse some relevant data:
energies = output.get_all_energies()
plt.plot(range(len(energies)), energies)
plt.scatter(range(len(energies)), energies)
```

Out[21]:
```
<matplotlib.collections.PathCollection at 0x12f63a510>
```

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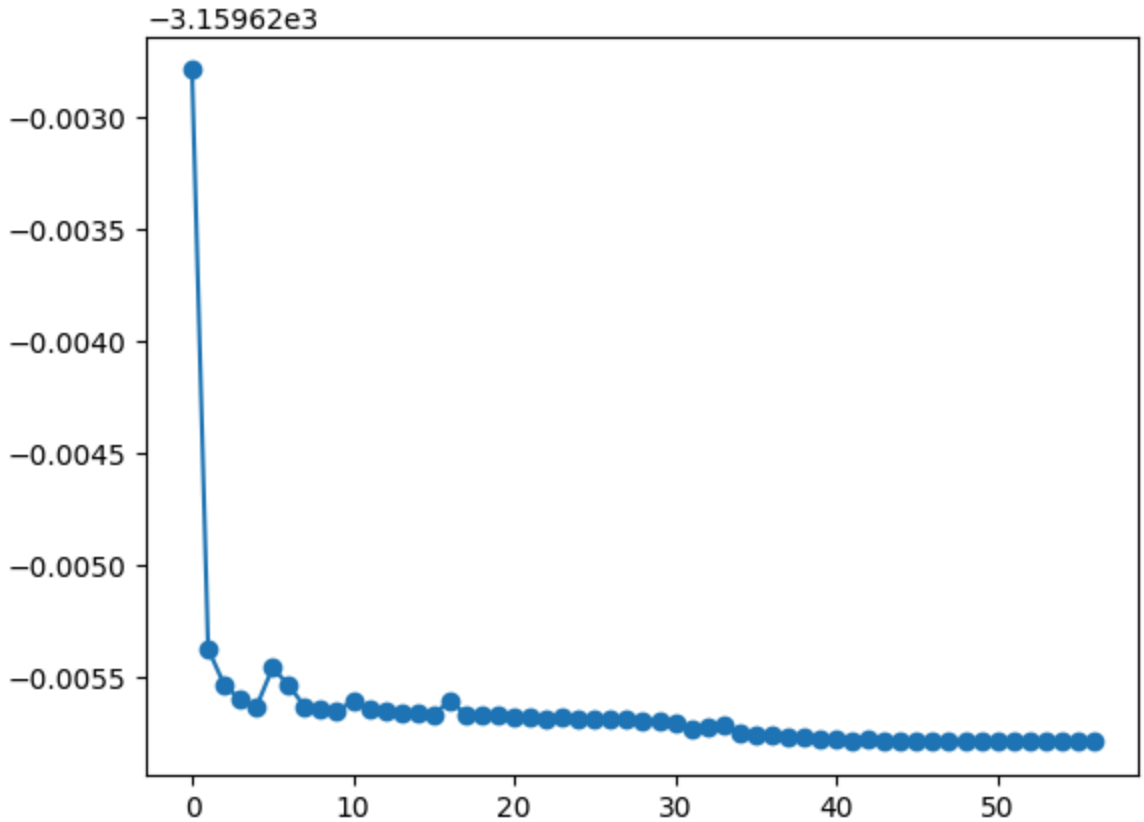

In [22]:
```
from scope.read_write import print_xyz
labels, coordinates = output.get_geometry_last_complete_block()
print_xyz(labels, coordinates)
```

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```
Fe  3.774774  3.779086  13.494815
S  1.627444  1.260647  16.645926
S  7.775067  6.070671  13.495128
N  4.388366  3.012781  11.573830
N  1.554286  4.139279  12.893164
N  3.225163  5.467599  14.748913
N  3.827655  5.637491  12.021693
N  3.234103  2.168291  14.532057
N  5.756036  4.156787  13.919598
C  4.403060  1.697532  11.314345
H  3.957564  1.056936  12.084132
C  4.962006  1.177906  10.147158
H  4.961604  0.098755  9.972590
C  5.513741  2.066869  9.219480
H  5.958902  1.697928  8.290368
C  5.487222  3.436157  9.493861
H  5.907017  4.161215  8.790346
C  4.922063  3.878771  10.694095
C  4.921934  5.340535  11.088554
H  5.883024  5.575005  11.579678
H  4.870551  5.973670  10.179266
C  0.533820  3.647869  13.609127
H  0.732976  2.747319  14.197467
C  0.147563  5.942191  12.164908
H  0.039650  6.860404  11.579998
C  1.366696  5.259855  12.173612
C  2.536976  5.694780  11.324614
H  2.590071  4.995985  10.471859
H  2.354671  6.703981  10.903917
C  2.671356  5.317313  15.960915
H  2.556322  4.289887  16.325468
C  2.247340  6.406355  16.722495
H  1.805775  6.242237  17.708857
C  2.394567  7.690638  16.190834
H  2.067764  8.569323  16.755395
C  2.959863  7.836761  14.922297
H  3.086086  8.825453  14.471378
C  3.374638  6.696879  14.225717
C  4.075222  6.796917  12.887107
H  3.786628  7.743812  12.387440
H  5.162524  6.856793  13.081945
C  2.572699  1.759360  15.438155
C  6.640933  4.935589  13.779979
C  -0.723383  4.258325  13.642042
H  -1.522564  3.818735  14.244730
C  -0.918572  5.431997  12.912247
H  -1.883471  5.948159  12.927194
```

How to Create a Computational Workflow Manually (Branch, Workflow, Job, Computation). Not recommended

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In this tutorial, we have navigated the computational workflow of an existing System (sys).

- sys had one branch ("*isolated*")
- isolated had two workflows ("*ref_hs_mol*" and "*ref_ls_mol*"), referring to two of the system's sources
- each workflow had three jobs ("*pbe scf*", "*pbe opt*", "*b3lyp opt*")
- each job had computations

This setup above was created automatically by SCOPE when running the workflow, based on the user's choice,
which are detailed in formatted input files that are explained in Tutorial 5.

You are **NOT** expected to handle the computational workflow **manually**. Indeed, it is **Not recommended**. However, it is possible:

Creating a Branch

In [23]:
```
## This is our system, with only the Isolated Branch (at least for now)
sys
```

Out[23]:
```
-------------------------------------
-- >>> SCOPE System_sco Object >>> --
-------------------------------------
 Version                = 1.0
 Type                   = system
 Subtype                = sco_system
 Source Path            = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources            = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches           = 1
     0: isolated, Num Workflows: 2
```

In [24]:
```
# Branches can be easily created. By default, a folder is also created
# This folder is where all computations associated with this branch would be
new_branch = sys.add_branch("Tutorial", create_folder=False)
```

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```
# If you run this cell twice, you'll get a message saying that the branch al
# The same branch (with the same name) cannot be created twice in the same s

## Here's the newly created branch
print(new_branch)

## When creating SOURCES, BRANCHES and WORKFLOWS, all names are de-capitaliz
## Notice that the name of the branch is now "tutorial" instead of "Tutorial
```

```
----------------------------------------------------
   >>> BRANCH
----------------------------------------------------
 System                = ABITEM
----------------------------------------------------
 self.status           = active
 self.creation_time    = 13/03/2026 16:53:46
 self.creation_user    = sergivela
 self.path             = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/tutorial/
 self.name             = tutorial
 Num Workflows         = 0
```

In [25]:

```
# Notice that a new Branch appears in the system
print(sys)
```

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```
-------------------------------------
-- >>> SCOPE System_sco Object >>> --
-------------------------------------
 Version               = 1.0
 Type                  = system
 Subtype               = sco_system
 Source Path           = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path     = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources           = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches          = 2
     0: isolated, Num Workflows: 2
     1: tutorial, Num Workflows: 0
```

Creating a Workflow

In [26]:
```python
# Similarly, workflows can be created inside a branch. Only the source name
new_workflow1 = new_branch.add_workflow("ref_hs_cell")
new_workflow2 = new_branch.add_workflow("ref_ls_cell")

print(new_workflow1)
```

```
----------------------------------------------------
   >>> >>> WORKFLOW
----------------------------------------------------
 Source Name                 = ref_hs_cell
 Source Type                 = cell
 Source sub-Type             = sco_cell
 Source Charge               = 0
 Source Spin                 = 16
 Source Phase                = HS
----------------------------------------------------
 Workflow Name               = ref_hs_cell
 Num Jobs                    = 0
```

Creating a Job

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In [27]:
```
# To create a Job, we need to provide the job_data file. For simplicity, we
new_job = new_workflow1.add_job(this_job.job_data)
```

In [28]:
```
# The new job should now appear in the workflow. Here the workflow:
new_workflow1
```

Out[28]:
```
----------------------------------------------------
   >>> >>> WORKFLOW
----------------------------------------------------
 Source Name                 = ref_hs_cell
 Source Type                 = cell
 Source sub-Type             = sco_cell
 Source Charge               = 0
 Source Spin                 = 16
 Source Phase                = HS
----------------------------------------------------
 Workflow Name               = ref_hs_cell
 Num Jobs                    = 1
        Last Job Name        = opt2
        Last Job Hierarchy   = 2
```

In [29]:
```
# and here the job:
new_job
```

Out[29]:
```
----------------------------------------------------
   >>> >>> >>> JOB
----------------------------------------------------
 Source Type           = cell
 Source Name           = ref_hs_cell
 Branch Name           = tutorial
 Workflow Name         = ref_hs_cell
----------------------------------------------------
 Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Computations/ABITEM/tutorial/
 Job name              = opt2
 Job type              = opt
 Job hierarchy         = 2
 Job requisites        = ['opt']
 Job constrains        = ['self']
 Job setup             = regular
 Num Computations      = 0
----------------------------------------------------
 self.isregistered (Temp) = False
 self.isgood       (Temp) = False
 self.isfinished   (Temp) = False
```

Creating a Computation

In [30]:
```
# To create a Computation, we need to provide the qc_data. Again, for simpli
new_comp = new_job.add_computation(this_comp.qc_data)

# Apart from that, computations need some more preparation:
# - This is to set the name of the computation files:
new_comp.set_name()
```

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```
new_comp.set_paths()
# - And this is to connect the computation with the initial and final states
new_comp.qc_data._add_attr("istate",new_job.istate)
new_comp.qc_data._add_attr("fstate",new_job.fstate)

# Alternatively, you can just use:
#new_job.set_computations_from_setup(this_comp.qc_data)

## This instruction will create 'all' computations expected from the job set
## I say 'all' instead of 'one' because some job setups require multiple com
```

In [31]:
```
# Eventually, the computation should appear in the job (see Num Computations
new_job
```

Out[31]:
```
----------------------------------------------------
   >>> >>> >>> JOB
----------------------------------------------------
 Source Type           = cell
 Source Name           = ref_hs_cell
 Branch Name           = tutorial
 Workflow Name         = ref_hs_cell
----------------------------------------------------
 Job path              = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Computations/ABITEM/tutorial/
 Job name              = opt2
 Job type              = opt
 Job hierarchy         = 2
 Job requisites        = ['opt']
 Job constrains        = ['self']
 Job setup             = regular
 Num Computations      = 1
-----------------------------------------------------
 self.isregistered (Temp) = False
 self.isgood       (Temp) = False
 self.isfinished   (Temp) = False
```

Resetting a Workflow

The objects associated with the computational workflow can be manually removed:

In [32]:
```
new_job.remove_computation(comp_keyword='', remove_files=False)
new_workflow1.remove_job(name='opt2')
new_branch.remove_workflow('ref_hs_cell')
sys.remove_branch("tutorial") ## or, alternatively: sys.remove_all_branches(
```

In [33]:
```
## And now, the branch is gone
sys
```

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Out[33]:
```
-------------------------------------
-- >>> SCOPE System_sco Object >>> --
-------------------------------------
 Version               = 1.0
 Type                  = system
 Subtype               = sco_system
 Source Path           = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path     = /Users/sergivela/Documents/SCOPE/Program/Scope_Tut
orials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources           = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches          = 1
     0: isolated, Num Workflows: 2
```

In [ ]:

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